

Inpatient Diabetes Cheat Sheet

UCSF endocrine modules (Rushakoff, Rao) · ADA 2022–23 era, verify drug details · Pre-op med/insulin dosing tool: [ucsf.logicnets.com](https://www.ucsf.edu/ucsf-logic-nets)

RULES

- **Goal BG 100–180**
- **No sliding scale alone** Basal-bolus = better BG + ~50% less hypo
- **T1DM / insulinopenic: always basal (or drip)** even if NPO → else DKA
- **Give SQ basal before stopping drip · Never stop pump w/o SQ basal or drip**
- Hold orals/injectables on admission (see bottom)

Starting doses

Patient	Total daily dose (TDD)	Split
Insulin-naïve	0.3–0.5 U/kg (0.3 = cautious; most end ≥0.5)	½ glargine · ½ aspart ÷ 3 meals
CKD · liver failure · s/p pancreatotomy	0.15–0.2 U/kg	½ · ½
Steroids	0.5 U/kg (up to >1) On insulin already → ↑ 30–50%	¼ glargine · ¾ aspart ÷ 3 meals
Home insulin, good control	70–80% of home dose Very high glargine → 50%	Same regimen
Home insulin, poor control	Re-dose by weight (above)	½ · ½
Home premix (70/30) → MDI	Same TDD	½ glargine · ½ aspart ÷ 3
IV drip → SQ	80% of last 24 h (steady rate) · post-CV surgery 60%	½ glargine · ½ aspart ÷ 3

- **Ex:** drip 1.5 U/h → 36 U × 0.8 ≈ 30 → glargine 15 + aspart 5 TID AC
- **Correction scale:** sensitive (BMI <25 / <50 U/d) · average (25–30 / 50–90) · resistant (>30 / >90)
- **Poor/unsure PO:** keep basal; aspart 0–50% or give right after meal by % eaten

Daily titration

FBG	<140	141–160	160–180	180–200	>200
Glargine	No change	+2–3 U	+4–5 U	+6–7 U	+8 U

- **Meal dose:** judge by next pre-meal BG (dinner → HS)
 - Flat → new dose = total given (meal + correction)
 - Rises → go higher than that
 - Low → ↓ that meal
- Don't split yesterday's correction evenly across doses. Glargine needs days → wait after big change

Steroids

- BG ↑ afternoon/evening, FBG often normal → **most insulin at lunch + dinner**
- Move correction scale up one level. Need persists while on steroid

NPO · tube feeds · TPN

- **Continuous TF / TPN / NPO on IVF → order "IP Adult SQ Insulin: NPO, TPN or Enteral Feedings"** (aspart q4h self-titrates to 121–180)
 - **Start aspart:** 2 U (no DM / <10 U/d) · 4 U (orals or <30 U/d) · 6 U (>30 U/d) · off drip = units given last 4 h
 - ± glargine 0.1–0.25 U/kg/d
 - **Not for:** cycled feeds, eating, frequent interruptions
- **Fixed-dose TF alt:** glargine ½ of 24 h drip total + aspart (total ÷ 10) q4h. No drip → 1 U per 6–10 g CHO/d
- **On insulin → NPO:** basal 70–80%. Orals: after procedure
- **Feeds held:** hold aspart, keep glargine. Unplanned stop >30 min → **D10W at feed rate**

Pumps

- Stay on if pt competent + has supplies + signs waiver + endo agrees. NPO for hours → may need drip
- Always consult endo

APeX insulin order sets

Order set	Use for
Adult SQ Insulin: Patient Eating	Premeal fixed · postmeal by amount eaten · CHO-counted pre/post
Adult SQ Insulin: NPO, TPN, Tube Feeding	q4h nutrition + correction · cycled TPN timing · automated q4h titration
Single Dose Insulin	One-off dose; BG checks + hypo orders time out
Adult Insulin Pump · ED Insulin Pump	Pump-specific orders
IV Insulin Protocol: ICU · Med/Surg	Initial rate by CVS / DKA / other
DKA: ER, ICU, Step-Down	IV + SQ options

Home non-insulin meds

Drug	Inpatient	Restart
Metformin	Hold	When stable, eGFR >30. No contrast hold needed

Drug	Inpatient	Restart
SU / meglitinide	Hold (hypo)	Only if eating reliably
DPP-4i	May continue (renal dose; not lina)	OK
GLP-1 RA	Hold (N/V)	Stable, no GI issue
SGLT2i	Hold. Stop 3–5 d pre-procedure . Post-op ↓CO2 → euDKA	When stable
TZD	Hold (fluid/CHF)	Discharge
Degludec	Lasts ~3 d; ↓20–30% starting 2–3 d pre-op; glargine sub may need less	Home dose

Insulins

Insulin	Onset	Peak	Duration	Give / check
Lispro · aspart · glulisine	15 min	0.5–1.5 h	3–5 h	10 min AC · check 2 h PP / next AC
Regular	30 min	2–5 h	6–8 h	30 min AC
NPH	1–2 h	4–12 h	18–26 h	AM / HS · check pre-dinner / FBG
Glargine U100 / U300	1.5 h	none	24 h	PM · check FBG
Degludec		none	>42 h	Anytime · FBG
Icodec (Awiqli) · efsitora (Onswik)	6 h	2–4 d	7+ d (weekly)	Anytime · FBG?

DKA / HHS

RULES

- Fluids → regular insulin 5 U/h (no bolus) → K
- BG <250 → add D5 + switch to IV insulin infusion order set (~2 U/h)
- Drip until HCO₃ ≥18 · SQ insulin before stopping drip
- Low K on arrival = huge deficit → replace before/with insulin

Orders (APeX DKA order set: ICU / ED / transitional only)

	Order
Fluids	NS 1 L/h × 4 h → 250–500 mL/h × 2–4 h → 100–250 mL/h (deficit ~100 mL/kg over 36–48 h) ½NS if corrected Na >150 · BG <250 → D5NS or D5½NS ~200 mL/h
Insulin	Regular 5 U/h IV, no bolus (order set titrates hourly). BG >600 → trust lab, not meter
K	>6: hold · 5–6: 10 mEq/h · 4–5: 20 · 3–4: 40 · <3: 40–80 mEq/h Max KCl 20 mEq/h central, 10 peripheral (module doesn't reconcile)
Labs	Chem q2h × 3 → q4h · BHB to follow ketones · I/O q2h
Phos / bicarb	Generally no. Bicarb only for fluid-refractory hypotension or hyperK arrhythmia (not pH alone)

UCSF order set vs ADA 2024 consensus (UpToDate)

	UCSF DKA order set	ADA 2024 / UpToDate
Insulin	Regular 5 U/h, no bolus	Regular 0.1 U/kg/h (bolus 0.1 U/kg only if drip delayed). Uncomplicated DKA: SQ rapid-acting 0.1 U/kg then 0.1 U/kg q1h or 0.2 q2h
Rate check	Order set titrates hourly	BG not ↓ 50–70 in 1st h → double. Max fall 90–120/h
BG <250	D5 ~200 mL/h, drip ~2 U/h	D5–10 ½NS 150–250 mL/h; drip 0.05 U/kg/h; keep BG 150–200 until resolved
K	Replace by level (above)	K <3.5: hold insulin , KCl 10–20 mEq/h until >3.5 · 3.5–5: 10–20 mEq/L of IVF · >5: none · K q2h
Fluids after bolus	Per protocol	Na nl/high → ½NS 250–500 mL/h · Na low → NS 250–500 mL/h
Bicarb	Not for pH alone	pH <7.0 → NaHCO ₃ 100 mEq in 400 mL SW over 2 h (+20 mEq KCl if K <5), repeat q2h until pH >7.0
Overlap	SQ before stopping drip	Drip 1–2 h after rapid-acting SQ; 2–4 h after basal. Insulin-naive TDD 0.5–0.6 U/kg

Transition to SQ

- HCO₃ ≥18 → SQ = **80% of last 24 h drip**, ½ glargine · ½ aspart ÷ meals (post-CV surgery 60%)
- Overlap: give SQ, then stop drip

Workup pearls

- **DKA criteria (ADA 2024):** BG ≥200 (or known DM) · pH <7.3 · HCO₃ <18 · BHB ≥3 or ketonuria ≥2+
- **Corrected Na** = Na + 2.4 × (BG – 100)/100
- Follow **BHB**, not urine ketones. ↑amylase, ↑WBC, ↑Cr can be spurious
- **SGLT2i + N/V/malaise** → **check ketones even if BG normal (euDKA)**
- Precipitants: missed insulin (incl. by us), infection, new DM, pancreatitis
- **HHS vs DKA:** BG >600, osm >350, pH + HCO₃ normal, ketones <5 · mortality 10–20% (DKA 3–10%)
- **HHS:** fluids are the treatment; start insulin after rehydration begun; find trigger (infection, MI/CVA, steroids)